

# Test Suite Architecture

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This document outlines the design principles and structural organization of our testing framework, which serves as the ultimate gate for ensuring visual precision and technical integrity across our layout parsing, cropping, and retrieval pipelines.

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## Core Principles

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### Strict Sandbox Isolation

To prevent tests from polluting our git history or modifying production databases, every test execution is strictly sandboxed. \* **Volatile Environments:** Tests operate inside a temporary directory created on-the-fly by the testing framework. \* **Global Redirection Safeguard:** The testing framework configures an automated, global environment-redirection safeguard (fully encapsulated via the global sandbox environment class) that executes on every single test case across the entire suite, preventing any un-sandboxed run from bypassing the barrier. \* **Dynamic Path Proxies:** To prevent early Python module imports from caching production path constants in-memory (which bypasses standard variable overrides), the database and search index directories are wrapped in dynamic proxies. These proxies evaluate the active repository root dynamically on every single access, permanently securing production folders from database or file leakage. \* **Zero Footprint:** Running tests leaves absolutely no transient artifacts, generated images, or log files in our production repository. \* **Strategic Transparency Exception:** While offline and deterministic tests leave zero footprint, our high-fidelity live visual agent integration tests selectively copy their generated visual outputs (such as discovered crops, coordinate overview grids, SQLite search indices, and execution log files) to a persistent test results folder. This explicit post-flight exfiltration grants human operators full transparency into the AI's actual real-world visual generations without compromising or bypassing the initial pipeline execution sandbox.

### Hybrid Fixture Strategy

We balance speed and real-world fidelity using two distinct types of data fixtures:

#### Dynamic Fixtures (Lightweight & Portable)

For fast, offline verification of coordinate scaling, layout bounding boxes, and name formatting, the framework programmatically draws a simple geometric PDF from scratch on every run. \* **Result:** This makes offline tests completely portable and self-contained, allowing anyone to check out the repository and immediately verify the codebase with zero binary download dependencies.

## Cached Fixtures (Real-World Fidelity)

To test visual classification, captioning, actual multimodal retrieval, and un-mocked rendering, the framework copies curated real-world document assets on-demand. Symmetrically, these are stored inside the local `.pytest_cache/fixtures/` directory, structured exactly as follows:

```
.pytest_cache/fixtures/
├── corpus/
│   └── 3048/
│       └── CAD-159825_Zusammenbau_3048.pdf          <-- Real CAD Drawing PDF!
├── library/
│   ├── fts.sqlite                                  <-- Pre-built BM25 SQLite Index!
│   ├── chroma/                                     <-- Pre-built Vector Cosine DB!
│   │   ├── chroma.sqlite3
│   │   └── ...
│   ├── layers.yaml                                <-- Layer Ontology!
│   ├── regions.yaml                               <-- Drafted Regions!
│   ├── violations.yaml                             <-- Authentic Violations!
│   ├── layer_distribution.svg
│   └── inventory/
│       ├── documents.yaml                          <-- Document Inventory!
│       ├── assets.yaml                             <-- Asset Inventory!
│       └── machine_catalog_3048.json                <-- Exported Catalog Schema!
└── crops/
    ├── drafts.yaml
    ├── 3048_CAD_159825_ZUSAMMENBAU_p01_isometrics_balloons.png <-- High-Res Detail Crop!
    ├── 3048_CAD_159825_ZUSAMMENBAU_p01_bom_table.png           <-- High-Res Macro Crop!
    └── 3048_BETRIEBSANLEITUNG_YUKONDAK_SI_p122_specs_table.png <-- High-Res Table Crop!
```

- **Recursive Corpus Sandboxing:** The synchronization fixture automatically detects our cached visual crops and real CAD schematics, recursively copying directories and safely routing loose PDF documents into their respective machine subfolders inside the volatile sandboxed tree.
  - **Strictly Read-Only:** The local cache serves exclusively as a read-only source repository; the framework reads these assets to populate the test sandbox but never writes active test run results back to it, preserving static input integrity.
  - **Selective Live Seeding:** When live integration tests execute, the framework selectively copies only the strictly required raw inputs (such as the layer ontology YAML and RTF batch questions) from the cached fixtures, allowing the visual discovery and verification loops to run un-mocked and generate their own outputs from scratch.
  - **Result:** This airtight grounding completely dismantles the "Mock-Comfort Trap"—allowing our unit, integration, and visual rendering tests to evaluate actual agentic page coordinate crops, OCR keyword extractions, JSON quotes repair, SQLite FTS5 BM25 queries, and ChromaDB vector similarity matching directly on real technical documents and databases, maintaining absolute data fidelity without any code mocks.
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## Execution Boundaries

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To preserve a rapid local development loop, our test suite is strictly categorized into three execution directories:

- **Unit Directory ( `test_suite/unit/` ):** Swift, pure algebraic/deterministic tests with zero network or database dependencies, completing in under 2 seconds.
- **Integration Directory ( `test_suite/integration/` ):** Tests verifying local PyMuPDF page rendering, FTS5 SQLite schema migrations, and coordinate voxel-grid rendering mathematical models offline.
- **Live Directory ( `test_suite/live/` ):** Live un-mocked API integration tests validating visual discovery agents, Slovenian and German technical translation mapping, conversational REPL loops, and multi-turn chat memory directly against the Gemini API.

To prevent visual intelligence and live integration tests from introducing pre-commit execution latency or unnecessary API token costs, all un-mocked visual tests are strictly decoupled and executed on-demand by explicit developer commands. Automated pre-commit hooks are configured to run strictly offline-only checks.

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# Unified Continuous Live Agent Validation

Rather than evaluating visual discovery and re-cropping agents in isolated, mocked test sessions, our framework implements a unified, end-to-end continuous live agent pipeline validation.

When executed on-demand, this pipeline runs sequentially inside a single, pristine test environment: 1. **Ingestion:** Parses Slovenian and German manuals and CAD drawing assets. 2. **Visual Discovery:** Launches the un-mocked discovery agent to explore technical pages and propose crop boundaries. 3. **Simulated Quality Failure:** Seeds a blank crop to trigger the verifier. 4. **Verification:** Launches the un-mocked verifier, which analyzes the crop and writes quality failure details. 5. **Refinement:** Launches the un-mocked re-cropping agent to refine the crop.

The entire output of this continuous sequence is preserved directly under the test results directory, providing human operators with full inspectability over the resulting folder structures.

## Test Catalog

Symmetrically, the test suite contains an ever-evolving, dynamically growing number of total test cases spanning our unit and integration pipelines. The suite consists of modular files targeting different functional domains of the technical catalog RAG pipeline. Below is the comprehensive index of our tests, their design purposes, and their execution capabilities:

### Test Catalog and Execution Matrix

| Test Concept Name                   | Offline | Live | Purpose / Target Verification                                                     |
|-------------------------------------|---------|------|-----------------------------------------------------------------------------------|
| API Connectivity Test               | [ ]     | [x]  | Validates basic Gemini API connectivity and API key existence.                    |
| Automated Translation Accuracy Test | [x]     | [x]  | Verifies automated language translation and multi-lingual technical translations. |

| Test Concept Name                     | Offline | Live | Purpose / Target Verification                                                                                               |
|---------------------------------------|---------|------|-----------------------------------------------------------------------------------------------------------------------------|
| Batch Query Execution Test            | [x]     | [x]  | Verifies batched sequence execution of search queries and citation compiling.                                               |
| Bbox Overlap Mathematics Test         | [x]     | [ ]  | Tests automated bbox checks and coordinate overlap math.                                                                    |
| Census Stats Aggregator Test          | [x]     | [ ]  | Validates checklist metrics and census stats calculations inside data inventories.                                          |
| Centralized Console Printing Test     | [x]     | [ ]  | Verifies standard printing, stderr routing, Rich style tag preservation, and bracket-swallowing safety filters dynamically. |
| CLI Documentation Sync Test           | [x]     | [ ]  | Validates synchronization between our codebase commands and markdown guides.                                                |
| CLI Retrieval Query Defaults Test     | [x]     | [ ]  | Verifies retrieval options, weights, and default thresholds used by the query CLI.                                          |
| Continuous E2E CLI Test               | [x]     | [x]  | End-to-end smoke test validating sequence runs across all pipeline CLI commands.                                            |
| Continuous Visual Agent Pipeline Test | [ ]     | [x]  | Continuous, un-mocked visual discovery, quality failure verification, and re-cropping.                                      |
| Coordinate Voxel-Grid Test            | [x]     | [ ]  | Verifies coordinate voxel-grid rendering for spatial layout mapping.                                                        |
| Corpus Directory Scanner Test         | [x]     | [ ]  | Verifies corpus structure, discovery and doc_id derivation rules.                                                           |
| Database Catalogue Exporter Test      | [x]     | [ ]  | Verifies final catalogue export filtering of approved inventory assets.                                                     |

| Test Concept Name                      | Offline | Live | Purpose / Target Verification                                                                                                                                           |
|----------------------------------------|---------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Database Migrations Schema Test        | [x]     | [ ]  | Validates FTS5 schema structures, SQLite database migrations, and index sanitization.                                                                                   |
| Deterministic Item State Trackers Test | [x]     | [ ]  | Verifies that <code>_track_item_state</code> thread-safely appends promoted and rejected census item identifiers to separate append-only tracker files in project logs. |
| Direct Page Fallback Pathway Test      | [x]     | [ ]  | Verifies the direct fallback pathway when a query is restricted by page and document.                                                                                   |
| Export Catalogue Schema Test           | [x]     | [ ]  | Validates structure and schema of machine catalogs exported as JSON.                                                                                                    |
| Extractor Prompt Normalization Test    | [x]     | [ ]  | Verifies string-escaping and structure normalization of extractor prompt blocks.                                                                                        |
| Filename Slug Resolution Test          | [x]     | [ ]  | Verifies that document filenames resolve to correct, uppercase, prefix-deduplicated slug identifiers.                                                                   |
| Images Agent Multi-Turn Discovery Test | [ ]     | [x]  | Validates multi-turn agentic loops for macro discovery directly against live endpoints.                                                                                 |
| Index Database Transaction Commit Test | [x]     | [ ]  | Verifies search indices database transaction logic and atomic commit operations.                                                                                        |
| Indexing Exclusions Test               | [x]     | [ ]  | Verifies indexing exclusions of TOC leaders and text-rich manuals.                                                                                                      |
| Indexing Write Protection Test         | [x]     | [ ]  | Verifies indexing safety boundaries, database write guards, and the <code>purge_indexes</code> rollback purger.                                                         |
| Interactive Document-Layer Gates Test  | [x]     | [ ]  | Verifies interactive human-in-the-loop (HITL) document layer assignment gates.                                                                                          |

| Test Concept Name                  | Offline                             | Live                                | Purpose / Target Verification                                                         |
|------------------------------------|-------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------|
| Inventory TOC Page Alignment Test  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies page-range alignment when mapping Table of Contents structures to documents. |
| Inventory YAML Schema Test         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Validates inventory document serialization, schemas, and metadata storage.            |
| Layer Classifier Test              | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies layer classification decisions from prompt contexts.                         |
| Layer Metadata Vectors Test        | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies indexing and retrieving small multi-class layer metadata vectors.            |
| Layer Proposer Ontology Test       | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Tests the layer proposer's initial layer ontology assignment loops.                   |
| Mangled Json Resilience Test       | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | Verifies pipeline actions using pre-extracted real-world manual samples.              |
| Manual Source Conflict Test        | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies detection of contradictions between conflicting manual sources.              |
| Multi-Machine Index Isolation Test | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies indexing isolation and query routing across multiple machines.               |
| Multilingual Term Patterns Test    | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies universal, Slovene, and German regex term patterns.                          |
| Page Layout Crop Test              | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies page layout crop rendering and filename collision-collapse.                  |
| Page Split Tiling Test             | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies grid overviews, page split-tiling, and JPEG preview resolutions.             |
| Prompt Template Compilation Test   | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies compilation and format of Gemini prompt templates for synthesis and citing.  |

| Test Concept Name                   | Offline                             | Live                                | Purpose / Target Verification                                                                                                                                                |
|-------------------------------------|-------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Query Routing Weight Test           | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies query-class classification (semantic vs. keyword) and weight blends.                                                                                                |
| Query Token Sanitization Test       | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies FTS5 query token escaping and SQL injection-guarding.                                                                                                               |
| Registry Inventory Splitter Test    | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies atomic splitting of inventory into documents and assets manifests.                                                                                                  |
| Resilient Census Registration Test  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies that <code>register_page_census</code> successfully normalizes and recovers double-nested, wrapped-list, and malformed raw string checklist payloads thread-safely. |
| Resilient Json Parsing Test         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies brace-counting JSON extraction and regex quote-repair.                                                                                                              |
| Resolution Bounds Safety Test       | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies resolution bounds, megapixels cap enforcement, and DPI safety.                                                                                                      |
| State Progressions Integration Test | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | Tests overall verifier flow and needs_verification flag status.                                                                                                              |
| Synonym Multi-Query Expansion Test  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies synonym generation and RRF-fused multi-query expansion.                                                                                                             |
| Table of Contents Parser Test       | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies table of contents (TOC) hierarchy building and page alignment.                                                                                                      |
| TOC Hierarchy Level Test            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies table of contents hierarchy parsing and heading level extraction.                                                                                                   |
| URN Provenance Hash Test            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies standard provenance-based URN generation and hashing.                                                                                                               |



| Test Concept Name                        | Offline                             | Live                                | Purpose / Target Verification                                                              |
|------------------------------------------|-------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------|
| Verifier Token Estimation Test           | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | Evaluates verifier tokens usage and spatial bounding box checks.                           |
| Visible Page Live Agent Interactive Test | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | Validates multi-turn visual re-cropping and verification loops against live API endpoints. |
| Visual Bounding Box Checker Test         | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies verifier automated checks and spatial region validations.                         |
| Visual Captioner Test                    | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Verifies generation of textual captions and vision prompts from region crops.              |